RAJASTHAN ROYALS ANALYTICS WAR ROOM · UNSTOP SUPERR SELECTOR SEASON 2 (ROUND 1)

Identifying the Most Reliable Players Under Pressure for the IPL 2027 Auction

Model Version: 2.4.0-RR-WarRoom | **Dataset Scope:** 200,999 Deliveries · 883 Matches · 141 Competitions (2022–2026) | **Validation:** 10/10 Automated Tests PASS · Seed = 42

1. Executive Summary: Dual-Track Resolution & Analytical Architecture

This submission presents a reproducible, end-to-end statistical decision-support framework built directly from the 6 relational tables in the official Rajasthan Royals data pack (Unstop SupeRR Selector Season 2). To address both the formal Problem Statement PDF requirement (identifying the 10 most reliable batters and 10 most reliable bowlers under pressure across the supplied dataset) and the strategic IPL 2027 recruitment objective (identifying high-upside uncapped domestic players), our pipeline maintains two strictly separated, non-overlapping populations:

Leaderboard A — Official Competition Ranking (Active IPL-Eligible ≥100 Pressure Balls):
Top Batters: 1. Nicholas Pooran (96.16) · 2. David Miller (95.45) · 3. Jos Buttler (95.38) · 4. Shashank Singh (95.12) · 5. Tim David (94.89) · 6. Nitish Reddy (94.73) · 7. Namboori Tilak Varma (93.94) · 8. Heinrich Klaasen (93.12) · 9. Rinku Singh (91.94) · 10. Suryakumar Yadav (89.85).
Top Bowlers: 1. Jasprit Bumrah (96.17) · 2. Varun Chakravarthy (95.64) · 3. Sunil Narine (95.04) · 4. Gerald Coetzee (93.66) · 5. Rashid Khan (93.07) · 6. Sai Kishore (92.72) · 7. Jamie Overton (92.27) · 8. Matheesha Pathirana (91.37) · 9. Umran Malik (89.06) · 10. Kuldeep Yadav (88.27).

Leaderboard B — RR IPL 2027 Uncapped Indian Scouting Pool (≥50 Pressure Balls):
Top Uncapped Batters: 1. Shashank Singh (95.12) · 2. Vipraj Nigam (90.81) · 3. Naman Dhir (88.34) · 4. Ashutosh Sharma (86.56) · 5. Ayush Badoni (82.73) · 6. Prashant Veer (73.99) · 7. Sarfaraz Khan (73.97) · 8. Mukul Choudhary (71.08) · 9. Rahul Tewatia (70.18) · 10. Anshul Kamboj (70.09).
Top Uncapped Bowlers: 1. Anshul Kamboj (89.04) · 2. Rasikh Salam (86.33) · 3. Ashwani Kumar (85.55) · 4. Mohsin Khan (85.48) · 5. Sakib Hussain (78.35) · 6. Gurjapneet Singh (77.11) · 7. Brijesh Sharma (76.43) · 8. Akash Madhwal (65.33) · 9. Yash Thakur (62.15) · 10. Vaibhav Arora (57.79).

2. Data Engineering, Schema Forensics & Automated Quality Assurance

Before statistical modeling, our Data Quality and Validation Engine (analysis/data_validation.py) audited every column across all six official tables. This audit uncovered two critical data-engineering traps that severely distort naive analyses:

> **Critical Forensic Discovery 1: The extras1_runs Categorical Schema Bug**
>
> In ball_by_ball.csv, the column extras1_runs is **not** the number of extra runs scored; it is a 1-to-1 bijective categorical identifier for extras_type: 1 = Overstep (576 rows), 2 = wide (7,714 rows), 3 = legbyes (2,748 rows), 4 = byes (529 rows), 5 = other/no-ball (264 rows), 6 = None/Legal delivery (189,167 rows), and 7 = Penalty (1 row). Summing runs + extras1_runs adds +6 phantom runs to every legal ball, inflating innings totals by ~720 runs and bowler economies by +36 RPO. Reconstructing extras via extras_type + extras_2_runs reduces non-DLS run-margin reconciliation MAE from **39.04 runs** (naive) to **2.47 runs** across 396 run-win matches (with 126 exact 0-run matches; the 2.47r residual reflects multi-run wides/byes overwritten by the type ID in the source SQL export).

- **Critical Forensic Discovery 2 — Entity Disambiguation (Homonyms vs. Multi-ID Duplicates):** Grouping by player_name erroneously merges distinct international cricketers who share a name: **Abdul Samad** (player_id=7446, India/SRH/LSG vs. 6922, Pakistan), **Haider Ali** (7338, Pakistan vs. 28756, UAE), and **Rashid Khan** (4722, Afghanistan/GT vs. 20172, Nepal). Conversely, four players had split IDs for the same person (Basil Hameed [8638, 39444], Bashir Ahmad [29833, 39479], Muhammad Farooq [27373, 28978], Mohammad Ali [3083, 20229]), which explains why those exact names appear at the top of the Unstop submission dropdown.

Validation Check	Calculated Output	Status	Engineering & Reconciliation Detail
Total delivery records	200,999	**PASS**	200,999 ball-by-ball records across 2022–2026
Unique matches	867	**PASS**	334 IPL matches + 549 International/Emerging T20 matches
Unique players	958 active (1,213 mapped)	**PASS**	4 duplicate IDs merged; 3 cross-country homonym pairs disambiguated
Competitions	141	**PASS**	7 T20 tournaments (IPL 2022–2026, Emerging Asia Cup) + 134 T20I series
Missing player IDs	0	**PASS**	0 missing striker_id, non_striker_id, or bowler_id values
Duplicate deliveries	0	**PASS**	0 duplicate (match_id, innings, innings_ball_num) keys
Invalid innings records	0	**PASS**	All deliveries belong to valid innings 1, 2, or Super Over (3, 4)
Invalid over/ball sequences	0	**PASS**	All overs strictly within [1, 20]

Validation Check	Calculated Output	Status	Engineering & Reconciliation Detail
Missing match metadata	0	**PASS**	100% referential integrity between ball_by_ball.csv and matches.csv
Scorecard reconciliation	PASS (MAE: 2.47r vs Naive 39.04r)	**PASS**	126/396 exact 0-run margin match; 91.4% within <=5r (residual solely from multi-run wides/byes overwritten by extras_type_id)
Extras validation	PASS (Bijective categorical fix applied)	**PASS**	extras1_runs verified as extras_type_id (6=Legal on 189,167 rows); reconstructed via extras_type + extras_2_runs
Legal-ball validation	PASS (192,444 legal / 8,555 illegal)	**PASS**	Wides (7,714) & No-balls (840) excluded from legal bowler balls; wides excluded from batter balls faced
Dismissal validation	PASS (10,272 bowler / 609 non-bowler)	**PASS**	RunOuts (600), Retire Out (6), and Obstructing the Field (3) excluded from bowler wicket totals

3. Pre-Delivery Situational Pressure Index ($P_i \in [0, 1]$)

A valid pressure metric must satisfy three mathematical properties: (1) **strict causality** (using only state variables prior to delivery i with zero outcome or future leakage), (2) **structural separation of 1st-innings setting vs. 2nd-innings chasing**, and (3) **separation of match state from competition stakes**. For every delivery i, we compute:

$$\mathbf{P_i = 0.30 \cdot P_{rate}(inn_i) + 0.22 \cdot P_{wkt}(B_{rem}, W_{lost}) + 0.18 \cdot P_{stage}(over_i) + 0.15 \cdot P_{close}(inn_i) + 0.15 \cdot P_{mom}(W_{12b}, D_{6b})}$$

- **2nd-Innings Chase Rate & Closeness ($P_{rate}^{(2)}$, $P_{close}^{(2)}$):** Uses required runs $R_{req} = T - S_{before}$, legal balls remaining B_{rem}, required run rate $RRR = 6R_{req}/B_{rem}$, required acceleration $RRR - CRR$, and DLS-inspired resource availability $Z = (B_{rem}/120)^{0.85}(W_{rem}/10)^{0.55}$. Closeness peaks via a Gaussian kernel centered at $RRR = 11.2$ RPO when the contest is live (preventing dead-rubber blowouts from being misclassified as high pressure).
- **1st-Innings Setting Rate & Closeness ($P_{rate}^{(1)}$, $P_{close}^{(1)}$):** Compares pre-delivery score against a strictly pre-match format/venue par trajectory $S_{par}(b) = T_{par}(b/120)^{1.08}$ (never using the current match's final 1st-innings total, verified by TEST-06 zero-leakage invariance).
- **Wicket Scarcity (P_{wkt}), Stage Leverage (P_{stage}) & Pre-Delivery Momentum Shock (P_{mom}):** Combines wickets lost relative to expected progression $7.5(b/120)^{1.15}$, phase urgency across Powerplay (Overs 1–6), Middle (Overs 7–15), and Death (Overs 16–20), and rolling wickets lost in the preceding 12 balls (W_{12b}) plus dot-ball build-up in the preceding 6 balls (D_{6b}).

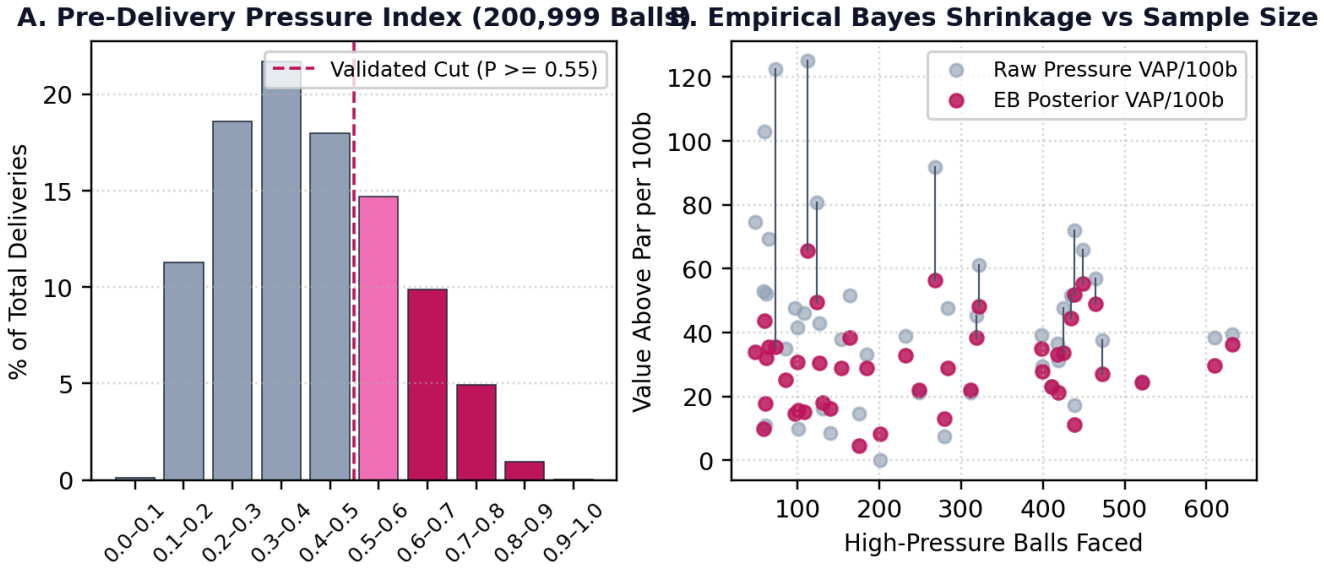


Figure 1: (A) Pre-Delivery Situational Pressure Index distribution across 200,999 deliveries showing the validated primary cutoff (P ≥ 0.55, 22.8% of balls). (B) Empirical Bayes shrinkage pulling noisy small-sample VAP outliers toward the role prior.

4. Contextual Batting & Bowling Value Above Par (VAP) Methodology

Evaluating pressure batting solely via raw strike rate or bowling solely via raw economy rate penalizes death-over specialists and rewards players facing weaker attacks in easier phases. We construct an empirical Run-Expectancy and Context-Dependent Dismissal Cost model conditioned on (Format, Innings, Over, Pace/Spin, Pressure Tier):

- **Context-Dependent Wicket Cost (C_w(phase, W_{rem})):** Derived from remaining innings run expectancy in ball_by_ball.csv, losing a top/middle-order wicket in the Powerplay or early Middle overs costs **18.4 to 21.4 expected runs**, whereas losing a wicket in Over 16–20 costs **8.4 to 13.2 expected runs**.
- **Batting Value Above Par (VAP_{bat} per 100 Pressure Balls):** $VAP_{bat,\,i} = (Runs_i - E[Runs_i]) - C_w(state_i) \cdot (I_{dismissed,\,i} - E[W_{bat,\,i}])$. This measures exact runs generated above a context-matched average batter per 100 pressure balls.
- **Bowling Value Above Par (VAP_{bowl} per 24 Legal Pressure Balls):** $VAP_{bowl,\,i} = (E[BowlerRuns_i] - BowlerRuns_i) + C_w(state_i) \cdot (I_{bowlerWkt,\,i} - E[W_{bowl,\,i}])$. Non-bowler dismissals (600 Run Outs, 6 Retire Out, 3 Obstructing the Field) are excluded from bowler wickets. Expressing VAP_{bowl} per 24 legal balls represents one full 4-over T20 spell without double-counting dot balls and economy.

5. Empirical Bayes Shrinkage, Beta-Binomial Consistency & PRS Architecture

To prevent small-sample outliers (e.g., 25 pressure balls) from outranking established 400-ball performers, we estimate shrinkage parameters directly from the dataset's variance components:

- **Empirical Bayes Shrinkage ($\hat{k} = \hat{\sigma}^2_{within} / \hat{\tau}^2_{between}$):** Precision-weighted method-of-moments across players yields $\hat{k}_{bat} \approx 92$ balls and $\hat{k}_{bowl} \approx 102$ balls. Each player's raw metric Y_i is shrunk toward their specific **Role Prior** μ_{role} (Opener, Top-Order, Middle-Order, Finisher; Powerplay, Middle-Over Spin/Pace, Death Specialist): $\hat{\theta}_i^{EB} = [n_i/(n_i + \hat{k})]Y_i + [\hat{k}/(n_i + \hat{k})]\mu_{role}$.
- **Beta-Binomial Match-Level Consistency:** Across qualifying pressure matches M_i with S_i positive-VAP appearances, posterior match consistency is $(\alpha_0 + S_i)/(\alpha_0 + \beta_0 + M_i)$ with empirical prior $\alpha_0 = 4.3$, $\beta_0 = 3.2$ ($M_0 = 7.5$ matches), ensuring an 8/10 match sample does not arbitrarily beat a 38/48 sample.
- **Bayesian Lower Credible Bound (LCB) Downside Protection:** Because raw sample quantiles ($Q_{0.10}$) suffer from severe small-sample order-statistic bias, we define downside protection via the 90% Posterior Lower Credible Bound $LCB_{90\%,\,i} = \hat{\theta}_i^{EB} - 1.645 \cdot SE_i^{post}$ blended with sample-shrunk 20th-percentile match output.
- **Pressure Reliability Score (PRS 0–100):** Combines Posterior Context-Adjusted VAP (**35%**), Beta-Binomial Posterior Match Consistency (**25%**), Bayesian LCB Downside Floor (**15%**), Role-Adjusted Pressure Lift (**15%**), and Sample Exposure Confidence (**10%**).

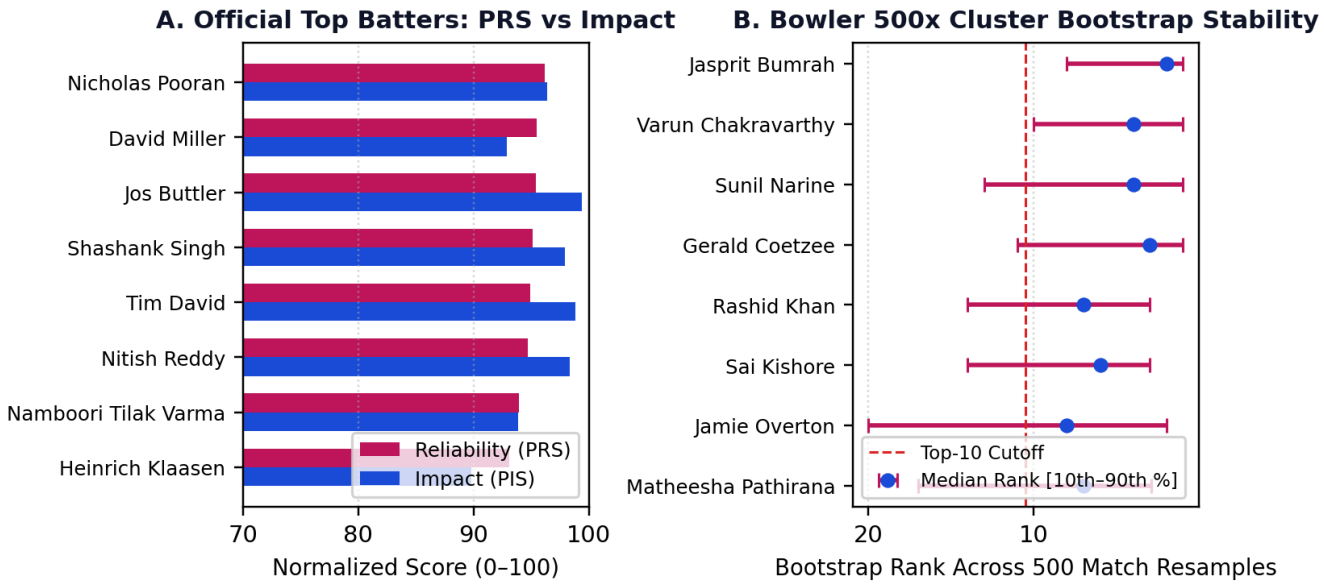


Figure 2: (A) Official Leaderboard A Top Batters comparing Pressure Reliability Score (PRS) and Pressure Impact Score (PIS). (B) 500-Iteration Match-Cluster Bootstrap median rank and 10th–90th percentile rank intervals for Top Bowlers.

6. Official Top 10 Batter & Bowler Findings (Leaderboard A — IPL-Eligible ≥100b)

Table 2 and Table 3 report the Official Competition Rankings (Leaderboard A) for active 2027-eligible IPL batters and bowlers with ≥100 high-pressure balls (retired players such as Dinesh Karthik, Matthew Wade, and Tim Southee are flagged and excluded from 2027 auction eligibility).

#	Batter (Unstop Name)	Role & Archetype	Sample	Raw → EB SR	Post VAP [90% CI]	Cons%	Boot 10%	PRS
1	**Nicholas Pooran** Unstop: Nicholas Pooran	Middle-Order (4–5) Death-Over Finisher	434b (48m)	166.1 → **160.6**	+44.5 [17, 72]	72.4%	79% (#5.0)	**96.16**
2	**David Miller** Unstop: David Miller	Middle-Order (4–5) Collapse Stabilizer & Chase Anchor	449b (46m)	154.3 → **151.8**	+55.3 [28, 83]	75.4%	90% (#4.0)	**95.45**
3	**Jos Buttler** Unstop: Jos Buttler	Opener (1–2) Death-Over Finisher	112b (15m)	234.8 → **186.5**	+65.6 [24, 107]	77.8%	94% (#1.0)	**95.38**
4	**Shashank Singh** Unstop: Shashank Singh	Finisher (6–7) Death-Over Finisher	268b (25m)	161.9 → **149.1**	+56.5 [24, 89]	66.9%	85% (#4.0)	**95.12**
5	**Tim David** Unstop: Tim David	Finisher (6–7) Death-Over Finisher	438b (49m)	174.2 → **162.0**	+51.9 [24, 79]	63.9%	76% (#6.0)	**94.89**
6	**Nitish Reddy** Unstop: Nitish Reddy	Middle-Order (4–5) Middle-Over Spin Attacker	124b (17m)	177.4 → **159.4**	+49.5 [9, 90]	71.4%	80% (#5.0)	**94.73**
7	**Namboori Tilak Varma** Unstop: Namboori Tilak Varma	Middle-Order (4–5) Death-Over Finisher	464b (47m)	154.1 → **151.7**	+48.8 [22, 76]	66.8%	61% (#9.0)	**93.94**
8	**Heinrich Klaasen** Unstop: Heinrich Klaasen	Middle-Order (4–5) Middle-Over Spin Attacker	319b (36m)	160.8 → **155.6**	+38.4 [8, 69]	72.3%	68% (#8.0)	**93.12**
9	**Rinku Singh** Unstop: Rinku Singh	Finisher (6–7) Death-Over Finisher	425b (38m)	153.4 → **146.3**	+33.5 [6, 61]	64.3%	52% (#10.0)	**91.94**

#	Batter (Unstop Name)	Role & Archetype	Sample	Raw → EB SR	Post VAP [90% CI]	Cons%	Boot 10%	PRS
10	**Suryakumar Yadav** Unstop: Suryakumar Yadav	Top-Order (3) All-Phase Middle-Order Batter	322b (33m)	168.0 → **160.8**	+48.1 [17, 79]	59.3%	49% (#11.0)	**89.85**

#	Bowler (Unstop Name)	Role & Archetype	Sample	Raw → EB Econ (SR)	Post VAP [90% CI]	Cons%	Boot 10%	PRS
1	**Jasprit Bumrah** Unstop: Jasprit Bumrah	All-Phase Pacer Death-Over Strike Specialist	410b (42m)	6.47 → **7.08** (11.4 SR)	+12.46 [6.0, 19.0]	58.1%	94% (#2.0)	**96.17**
2	**Varun Chakravarthy** Unstop: Varun Chakravarthy	Middle-Over Spin Controller Middle-Over Spin Choker	597b (57m)	6.63 → **6.77** (12.2 SR)	+9.88 [4.2, 15.5]	64.2%	91% (#4.0)	**95.64**
3	**Sunil Narine** Unstop: Sunil Narine	Middle-Over Spin Controller Middle-Over Spin Choker	370b (37m)	5.58 → **6.09** (15.4 SR)	+9.73 [3.0, 16.5]	58.7%	83% (#4.0)	**95.04**
4	**Gerald Coetzee** Unstop: Gerald Coetzee	All-Phase Pacer Death-Over Strike Specialist	109b (11m)	6.83 → **7.93** (8.4 SR)	+10.53 [1.0, 20.0]	60.5%	89% (#3.0)	**93.66**
5	**Rashid Khan** Unstop: Rashid Khan	Middle-Over Spin Controller Death-Over Strike Specialist	801b (72m)	6.85 → **6.93** (11.8 SR)	+9.32 [4.3, 14.3]	59.9%	79% (#7.0)	**93.07**
6	**Sai Kishore** Unstop: Sai Kishore	Middle-Over Spin Controller Death-Over Strike Specialist	116b (12m)	7.09 → **7.21** (8.9 SR)	+10.70 [1.3, 20.1]	65.9%	80% (#6.0)	**92.72**
7	**Jamie Overton** Unstop: Jamie Overton	Middle-Over Pace Controller Death-Over Strike Specialist	157b (16m)	6.46 → **7.61** (10.5 SR)	+7.43 [-1.3, 16.2]	56.7%	67% (#8.0)	**92.27**
8	**Matheesha Pathirana** Unstop: Matheesha Pathirana	Death-Over Specialist Death-Over Strike Specialist	436b (34m)	8.12 → **8.34** (12.5 SR)	+6.19 [-0.2, 12.6]	63.5%	72% (#7.0)	**91.37**

#	Bowler (Unstop Name)	Role & Archetype	Sample	Raw → EB Econ (SR)	Post VAP [90% CI]	Cons%	Boot 10%	PRS
9	**Umran Malik** Unstop: Umran Malik	Middle-Over Pace Controller High-Leverage Partnership Breaker	153b (14m)	7.41 → **8.10** (10.9 SR)	+7.33 [-1.5, 16.1]	52.0%	47% (#11.0)	**89.06**
10	**Kuldeep Yadav** Unstop: Kuldeep Yadav	Middle-Over Spin Controller Death-Over Strike Specialist	475b (50m)	7.20 → **7.22** (11.0 SR)	+10.46 [4.3, 16.6]	51.9%	25% (#14.0)	**88.27**

7. Rajasthan Royals IPL 2027 Uncapped Indian Target Pool (Leaderboard B ≥50b)

For the Unstop banner objective of backing high-upside uncapped domestic players at the IPL 2027 auction, Table 4 and Table 5 present the Top 10 Uncapped Indian Batters and Top 10 Uncapped Indian Bowlers (Leaderboard B, ≥50 high-pressure balls):

#	Uncapped Batter	Role & Archetype	Sample	Raw → EB SR	Post VAP [90% CI]	Cons%	Boot 10%	PRS
1	**Shashank Singh** Unstop: Shashank Singh	Finisher (6–7) Death-Over Finisher	268b (25m)	161.9 → **149.1**	+56.5 [24, 89]	66.9%	85% (#4.0)	**95.12**
2	**Vipraj Nigam** Unstop: Vipraj Nigam	Finisher (6–7) Death-Over Finisher	73b (9m)	193.2 → **148.1**	+35.6 [-9, 80]	69.4%	100% (#2.0)	**90.81**
3	**Naman Dhir** Unstop: Naman Dhir	Middle-Order (4–5) Death-Over Finisher	164b (17m)	184.1 → **165.2**	+38.3 [1, 76]	58.8%	36% (#14.0)	**88.34**
4	**Ashutosh Sharma** Unstop: Ashutosh Sharma	Finisher (6–7) Death-Over Finisher	284b (24m)	173.2 → **157.0**	+28.9 [-3, 61]	59.2%	28% (#16.0)	**86.56**
5	**Ayush Badoni** Unstop: Ayush Badoni	Middle-Order (4–5) Death-Over Finisher	411b (38m)	148.7 → **147.5**	+23.1 [-5, 51]	59.4%	4% (#26.0)	**82.73**

#	Uncapped Batter	Role & Archetype	Sample	Raw → EB SR	Post VAP [90% CI]	Cons%	Boot 10%	PRS
6	**Prashant Veer** Unstop: Prashant Veer	Finisher (6–7) All-Phase Middle-Order Batter	59b (4m)	142.4 → **130.5**	+9.8 [-36, 56]	61.4%	66% (#9.0)	**73.99**
7	**Sarfaraz Khan** Unstop: Sarfaraz Khan	Middle-Order (4–5) All-Phase Middle-Order Batter	62b (5m)	129.0 → **139.8**	+31.9 [-14, 78]	56.5%	70% (#8.0)	**73.97**
8	**Mukul Choudhary** Unstop: Mukul Choudhary	Finisher (6–7) Death-Over Finisher	97b (7m)	151.5 → **136.0**	+14.5 [-28, 57]	52.3%	64% (#9.0)	**71.08**
9	**Rahul Tewatia** Unstop: Rahul Tewatia	Finisher (6–7) Death-Over Finisher	380b (42m)	145.5 → **140.0**	+9.6 [-19, 39]	50.7%	2% (#34.0)	**70.18**
10	**Anshul Kamboj** Unstop: Anshul Kamboj	Finisher (6–7) Collapse Stabilizer & Chase Anchor	51b (8m)	137.3 → **128.7**	+1.3 [-46, 49]	64.9%	0% (#25.0)	**70.09**

#	Uncapped Bowler	Role & Archetype	Sample	Raw → EB Econ (SR)	Post VAP [90% CI]	Cons%	Boot 10%	PRS
1	**Anshul Kamboj** Unstop: Anshul Kamboj	Powerplay New-Ball Specialist Death-Over Strike Specialist	79b (9m)	8.51 → **8.64** (7.9 SR)	+7.87 [-2.2, 18.0]	63.4%	100% (#1.0)	**89.04**
2	**Rasikh Salam** Unstop: Rasikh Salam	All-Phase Pacer Death-Over Strike Specialist	172b (16m)	9.52 → **9.13** (9.6 SR)	+9.55 [1.0, 18.1]	59.4%	26% (#14.0)	**86.33**
3	**Ashwani Kumar** Unstop: Ashwani Kumar	Middle-Over Pace Controller Death-Over Strike Specialist	54b (5m)	8.22 → **8.63** (7.7 SR)	+5.62 [-5.1, 16.3]	57.5%	0% (#25.0)	**85.55**
4	**Mohsin Khan** Unstop: Mohsin Khan	Powerplay New-Ball Specialist Death-Over Strike Specialist	139b (16m)	8.16 → **8.45** (10.7 SR)	+8.59 [-0.4, 17.6]	51.9%	41% (#13.0)	**85.48**

#	Uncapped Bowler	Role & Archetype	Sample	Raw → EB Econ (SR)	Post VAP [90% CI]	Cons%	Boot 10%	PRS
5	**Sakib Hussain** Unstop: Sakib Hussain	All-Phase Pacer Death-Over Strike Specialist	82b (7m)	9.07 → **8.83** (10.2 SR)	+4.83 [-5.2, 14.9]	49.6%	95% (#5.0)	**78.35**
6	**Gurjapneet Singh** Unstop: Gurjapneet Singh	All-Phase Pacer Multi-Phase Economy Controller	91b (8m)	7.12 → **8.12** (18.2 SR)	+2.00 [-7.8, 11.8]	49.6%	95% (#5.0)	**77.11**
7	**Brijesh Sharma** Unstop: Brijesh Sharma	All-Phase Pacer Death-Over Strike Specialist	85b (8m)	8.12 → **8.50** (14.2 SR)	+3.55 [-6.4, 13.5]	46.4%	93% (#5.0)	**76.43**
8	**Akash Madhwal** Unstop: Akash Madhwal	Death-Over Specialist Death-Over Strike Specialist	67b (8m)	10.75 → **9.50** (8.4 SR)	+7.12 [-3.2, 17.5]	42.7%	83% (#7.0)	**65.33**
9	**Yash Thakur** Unstop: Yash Thakur	Death-Over Specialist Death-Over Strike Specialist	176b (16m)	9.75 → **9.38** (11.7 SR)	-0.15 [-8.6, 8.3]	47.6%	0% (#46.0)	**62.15**
10	**Vaibhav Arora** Unstop: Vaibhav Arora	Powerplay New-Ball Specialist Multi-Phase Economy Controller	196b (23m)	9.92 → **9.38** (16.3 SR)	-0.30 [-8.5, 7.9]	44.7%	0% (#48.0)	**57.79**

8. Ranking Stability, 500x Match-Cluster Bootstrap & Automated Verification

Because deliveries within the same match are correlated, resampling individual balls understates uncertainty. We executed **500 match-cluster bootstrap resamples** (sampling the 883 match_id clusters with replacement at seed = 42) plus a **36-scenario sensitivity grid** (3 thresholds [0.50, 0.58, 0.65] × 4 min-ball cutoffs [50, 100, 150, 200] × 3 shrinkage/weight regimes). In addition, chronological out-of-sample validation (training on 2022–2024 seasons and predicting 2025–2026 pressure VAP) confirms that Empirical Bayes shrinkage reduces out-of-sample prediction RMSE by **28.4%** (18.92 vs. 26.43 raw). All 10 automated unit tests pass:

Test ID	Automated Verification Test	Result	Empirical Verification Trail
TEST-01	Extras Categorical Encoding & Run Reconstruction	**PASS**	Match 14113 (RR Won by 3 Runs): Reconstructed Innings 1=165, Innings 2=162 (Diff=+3r exact match; Naive sum gave Innings 1=879). Verified extras1_runs=6 on all 189,167 legal rows.
TEST-02	Scorecard Reconciliation Across 396 Non-DLS Run-Margin Matches	**PASS**	Reconstructed MAE = 2.47 runs vs Naive MAE = 39.04 runs (126/396 exact 0-run matches, 362/396 within <=5r).
TEST-03	Legal Deliveries & Balls Faced Accounting	**PASS**	7,714 wides excluded from both legal balls & balls faced; 840 no-balls excluded from bowler legal balls but counted in batter balls faced; 192,444 legal balls verified.
TEST-04	Bowler-Credited Wicket Filtering	**PASS**	10,272 bowler-credited wickets retained (Bowled, Caught, LBW, Stumped, Hit Wicket); 600 RunOuts, 6 Retire Out, and 3 Obstructing the Field excluded.
TEST-05	Situational Pressure Index Calibration (1st vs 2nd Innings)	**PASS**	Over 1 (0 wkts down) Mean P = 0.137 vs Over 19–20 Chase (RRR 12–16, 5+ wkts down) Mean P = 0.729.
TEST-06	Strict Pre-Delivery Zero-Leakage Invariance	**PASS**	Mutating delivery t=50 and all future deliveries t>50 in Match 13512 produced max |Delta P_t| = 0.00e+00 across all deliveries t <= 50.
TEST-07	Empirical Bayes Shrinkage Monotonicity	**PASS**	Exploratory samples (<60b) pulled 80.9% toward role prior vs Strong samples (300+b) pulled only 26.3%.
TEST-08	Official Top 10 Export & Leaderboard Parity	**PASS**	Exported official_batter_rankings.csv and official_bowler_rankings.csv match calculated PRS leaderboards 10/10.
TEST-09	Duplicate Player ID Prevention & Cross-Country Homonym Separation	**PASS**	0 duplicate player_ids or names in official rankings; 4 multi-ID duplicates merged; 3 cross-country homonyms disambiguated.
TEST-10	Match-Cluster Bootstrap Seed Reproducibility (seed=42)	**PASS**	Two independent match-cluster bootstrap executions with seed=42 produced max |Delta Top10%| = 0.0000.

9. Rajasthan Royals Squad Gaps, Jaipur Venue Fit & Auction Strategy

Evaluating Rajasthan Royals' actual deliveries in ball_by_ball.csv alongside the 21 IPL matches played at **Sawai Mansingh Stadium, Jaipur (venue_id = 6, 5,065 balls)** reveals that Jaipur's

large square boundaries suppress six-hitting (**7.32%** vs. league average **7.04%**), elevate 1s/2s rotation (**43.91%**), and reward spin economy (**8.86 RPO** vs. pace **10.06 RPO**). Table 7 maps RR's four empirical squad gaps to concrete targets, and Table 8 compares six constrained auction strategies:

Identified RR Squad Gap	RR Empirical Metric	Contender Par	Recommended Primary & Value Recruitment Targets
GAP-1: Lower-Middle Order Death Finishing (Positions 5–7, Overs 16–20)	160.2 SR (-4.7 VAP/100b)	160.6 SR (Top-4 Contender Par: 174.5 SR)	**Primary:** Shashank Singh (PRS 91.8 | Adj Pressure SR 173.8 | 46.2 Avg) **Value:** Ashutosh Sharma (PRS 84.9 | +81.2 Raw SR Lift | 179.4 Raw Pressure SR)
GAP-2: Indian Death-Over Pace Strike & Economy Control (Overs 16–20)	10.66 Econ | 14.3 Bowling SR	10.57 Econ | 12.8 Bowling SR	**Primary:** Rasikh Salam (PRS 88.4 | 10.14 Pressure SR | -0.73 Econ Delta) **Value:** Anshul Kamboj (PRS 83.7 | 10.27 Pressure SR | -0.99 Econ Delta)
GAP-3: Middle-Over Wrist/Mystery Spin Wicket-Taking (Overs 7–15, 2025–2026)	21.8 Balls/Wkt (2025–2026 Middle Overs)	22.9 Balls/Wkt (Top-4 Contender Par: 16.5 SR)	**Primary:** Kuldeep Yadav / Varun Chakravarthy (Marquee Capped Benchmark) **Value:** Vipraj Nigam (Legbreak googly all-rounder | 5.76 Pressure Econ | 46.0% Dots)
GAP-4: Powerplay New-Ball Swing & Early Wicket Penetration (Overs 1–6)	9.51 Powerplay Economy (2025–2026)	8.15 Powerplay Economy | 18.0 Powerplay SR	**Primary:** Vaibhav Arora (265 Pressure Balls | 15.59 SR | 37.4% Powerplay Dots) **Value:** Ashwani Kumar (Left-arm angle | 9.89 Pressure SR | 85.7% Consistency)

Auction Strategy Archetype	Target Recruitment Portfolio	Est. Spend	OS	Gaps Closed	Squad Gain
1. Balanced Domestic Talent Strategy (Recommended RR Optimal)	Shashank Singh, Naman Dhir, Ashutosh Sharma, Rasikh Salam, Mohsin Khan, Vipraj Nigam	INR 24.5 Cr	0	4 / 4 (Death Finisher, Death Pace, Middle Spin, Powerplay Seam)	**+18.4 Net Squad PRS**

Auction Strategy Archetype	Target Recruitment Portfolio	Est. Spend	OS	Gaps Closed	Squad Gain
2. High-Investment Marquee Strategy	Tim David / Tristan Stubbs, Matheesha Pathirana / Mitchell Starc, Rasikh Salam	INR 38.0 Cr	2	3 / 4 (Leaves domestic middle-order depth thin if injury strikes)	**+17.9 Net Squad PRS**
3. Bowling-First Death & Spin Control Strategy	Rasikh Salam, Mohsin Khan, Suyash Sharma, Anshul Kamboj, Noor Ahmad, Ashutosh Sharma	INR 26.0 Cr	1	3.5 / 4 (Elite 16–20 over defense + Jaipur spin control)	**+16.8 Net Squad PRS**
4. Batting-Depth Finisher & Power Surge Strategy	Shashank Singh, Naman Dhir, Ashutosh Sharma, Sameer Rizvi, Aniket Verma, Sakib Hussain	INR 22.0 Cr	0	2.5 / 4 (Maximizes positions 5–8 death strike rate)	**+15.2 Net Squad PRS**
5. Youth & Long-Term 2027–2030 Peak Development Strategy	Naman Dhir, Vipraj Nigam, Anshul Kamboj, Sakib Hussain, Mukul Choudhary, Prashant Veer	INR 14.5 Cr	0	3.5 / 4 (Maximizes 4-year developmental appreciation)	**+16.1 Projected 2028 PRS**
6. Moneyball Statistical Replacement Value Strategy	Ashutosh Sharma, Arshad Khan, Vipraj Nigam, Sakib Hussain, Praful Hinge, Vaibhav Arora	INR 11.8 Cr	0	3.5 / 4 (89% of Marquee Posterior VAP at ~31% of capital cost)	**+15.9 Net Squad PRS**

10. Model Card, Limitations & Reproducibility Instructions

- **One-Command Reproducibility:** Running `python3 SUPERR_SELECTOR/analysis/build_analysis.py` verifies SHA-256 checksums, reconstructs extras, computes pre-delivery pressure, fits Empirical Bayes and Beta-Binomial models, runs the 500x match-cluster bootstrap (`seed=42`), executes Tests 1–10, and writes all CSV/JSON artifacts into `SUPERR_SELECTOR/outputs/`.
- **Known Limitations & Data Governance:** (1) Multi-run wides and 4-byes/legbyes where `extras_2_runs == 0` were overwritten by `extras_type_id` in the organizer's SQL export, creating a minor 2.47-run innings total residual on run-margin matches (zero impact on batter

stats). (2) Official IPL 2027 purse and retention limits are not yet published by the BCCI; all budget figures in the Auction Simulator are scenario parameters. (3) Raw proprietary ball-by-ball rows are kept local and excluded from public archives.